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Ecole Nationale Superieure d'Informatique (ESI)

Advanced Machine Learning
Module: Federated Learning and Transfer Learning

Personalization in Federated Learning under Domain Shift

Final Project Report

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Abstract

Federated Learning (FL) enables several institutions to train machine learning models without centralizing private data. In healthcare, however, hospitals often have different patient populations and label distributions. This domain shift can make a single global model unreliable for individual centers.

This project studies personalization in FL using the FLamby Heart Disease dataset, which contains 920 samples distributed across four medical centers. We compare local-only learning, FedAvg, and pFedMe. FedAvg reaches an average accuracy of 0.775, while pFedMe reaches 0.835 and matches the local-only baseline. The strongest failure case appears on the Switzerland center, where FedAvg drops to 0.640 while pFedMe reaches 0.840. The results show that personalization is useful under non-IID healthcare data, while larger validation, privacy controls, and communication-aware experiments remain necessary before deployment.

1 Introduction

1.1 Motivation

Healthcare data is naturally distributed across hospitals and medical centers. Federated Learning is attractive because it allows institutions to collaborate without directly sharing patient records. Standard FL methods such as FedAvg train one global model, but this can fail when client data distributions are very different. Personalized FL addresses this issue by allowing each client to adapt the shared model to its local domain.

1.2 Objectives

The objectives of this project are:

- implement and compare local-only learning, FedAvg, and pFedMe;
- analyze the effect of domain shift on each medical center;
- study the personalization parameter λ;
- provide a reproducible and critically discussed experimental workflow.

2 Dataset and Exploratory Analysis

2.1 Dataset Description

The experiments use the FLamby Heart Disease dataset. The binary task is to predict the presence of heart disease from 13 clinical features. The dataset contains 920 samples from Cleveland, Hungary, Switzerland, and LongBeach.

Table 1: Dataset distribution across centers.

Center	Samples	Train	Test	Positive rate (train)
Cleveland	303	242	61	0.459
Hungary	294	235	59	0.362
Switzerland	123	98	25	0.939
LongBeach	200	160	40	0.744
Total	**920**	**735**	**185**	–

2.2 Domain Shift

The main challenge is the difference in class distributions between centers. Switzerland has a very high disease-positive rate, while Hungary has the lowest. This makes the problem strongly non-IID and explains why a single global model can struggle.

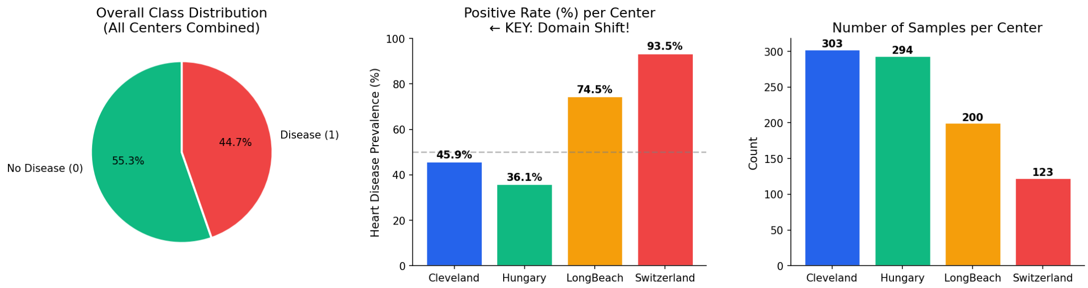

Figure 1: Class distribution by center. The positive-rate gap between centers is the main source of domain shift.

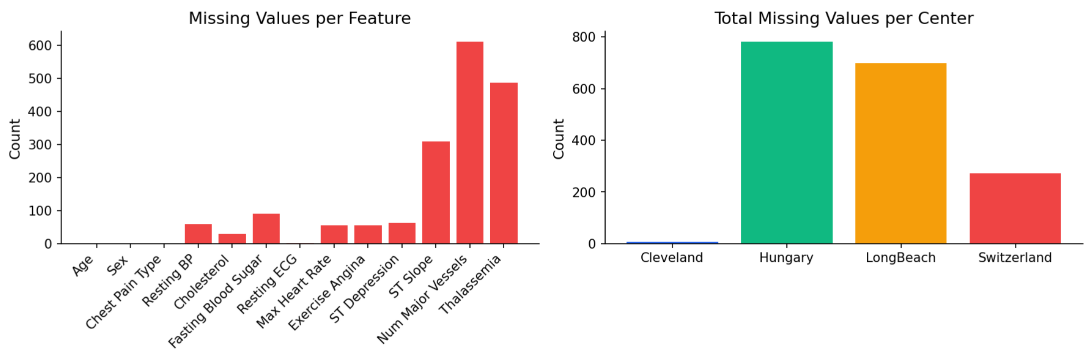

Figure 2: Missing-value pattern across clinical features and centers.

3 Methodology

3.1 Compared Methods

Table 2: Compared learning strategies.

Method	Training idea	Personalization	Expected behavior
Local-only	Each center trains independently.	Full local adaptation.	Strong local fit, no collaboration.
FedAvg	Clients train locally and the server averages weights.	None.	Collaborative but sensitive to domain shift.
pFedMe	Clients optimize a local objective regularized toward a global model.	Controlled by λ.	Balances shared knowledge and local adaptation.

3.2 Experimental Protocol

All methods use the same train/test split and preprocessing pipeline. Missing values are imputed using training data only, and features are standardized using training statistics. The classifier is logistic regression with balanced class weights. FedAvg and pFedMe are trained for 20 communication rounds. The ablation evaluates $\lambda \in \{0.0, 0.1, 0.5, 1.0\}$.

4 Results

4.1 Main Results

Table 3: Average performance across centers. Values are mean ± standard deviation.

Method	Accuracy	Precision	Recall	F1-score
Local-only	0.835 ± 0.051	0.850 ± 0.050	0.878 ± 0.065	0.861 ± 0.034
FedAvg	0.775 ± 0.082	0.808 ± 0.069	0.850 ± 0.100	0.820 ± 0.027
pFedMe ($\lambda = 0.5$)	0.835 ± 0.051	0.850 ± 0.050	0.878 ± 0.065	0.861 ± 0.034

pFedMe improves average accuracy by 0.060 over FedAvg and matches the local-only baseline in this run. This matters because pFedMe keeps a federated structure while avoiding the worst behavior of the global model.

4.2 Per-Center Performance

Table 4: Per-center accuracy comparison.

Center	Samples	Positive rate	Local-only	FedAvg	pFedMe
Cleveland	303	0.459	0.869	0.836	0.869
Hungary	294	0.362	0.881	0.847	0.881
Switzerland	123	0.939	0.840	0.640	0.840
LongBeach	200	0.744	0.750	0.775	0.750

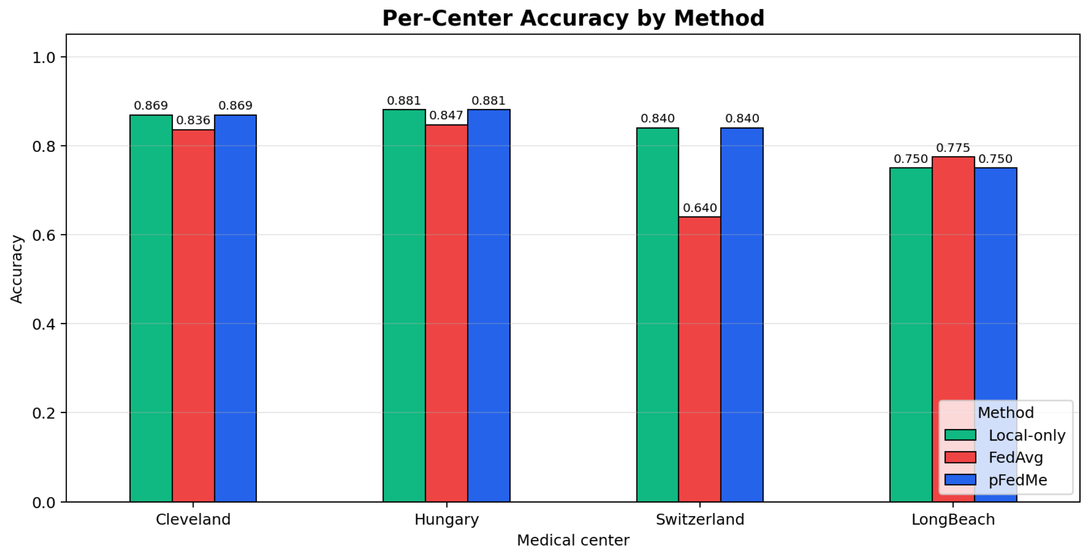

Figure 3: Per-center accuracy by method. Switzerland is the clearest FedAvg failure case.

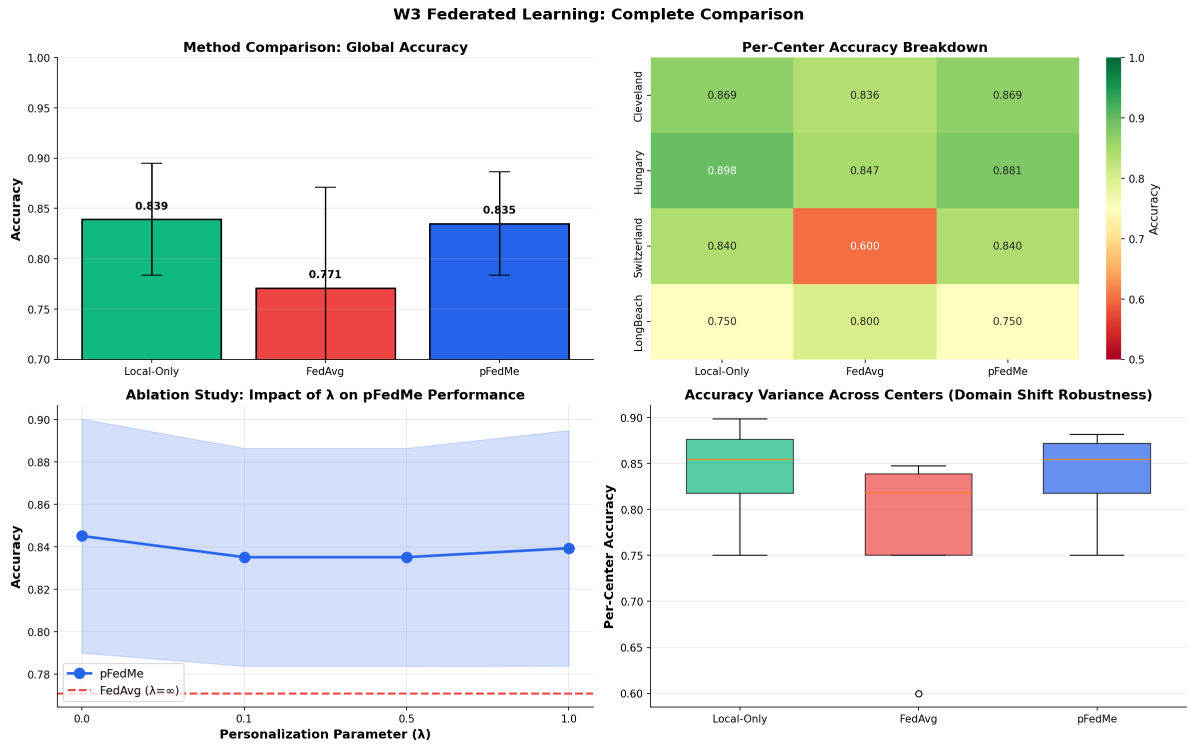

Figure 4: W3 federated learning comparison and ablation visualizations.

4.3 Ablation and Statistical Diagnostic

Table 5: pFedMe personalization-parameter ablation.

λ	**Accuracy**	**F1-score**	**Interpretation**
0.0	0.845 ± 0.055	0.867 ± 0.043	Highest raw accuracy in this run.
0.1	0.835 ± 0.051	0.861 ± 0.034	Slight personalization.
0.5	0.835 ± 0.051	0.861 ± 0.034	Main configuration; stable trade-off.
1.0	0.835 ± 0.051	0.861 ± 0.034	Stronger local adaptation.

The paired permutation diagnostic over four centers gives a pFedMe–FedAvg mean accuracy delta of 0.0605 with a two-sided p-value of 0.25. Because only four centers are available, this should be interpreted as directional evidence rather than a strong statistical significance claim.

5 Critical Analysis

FedAvg is vulnerable when one global model must serve centers with very different distributions. The Switzerland center is the strongest example: its positive rate is much higher than the others, and FedAvg accuracy falls to 0.640. pFedMe recovers to 0.840 by allowing center-specific adaptation.

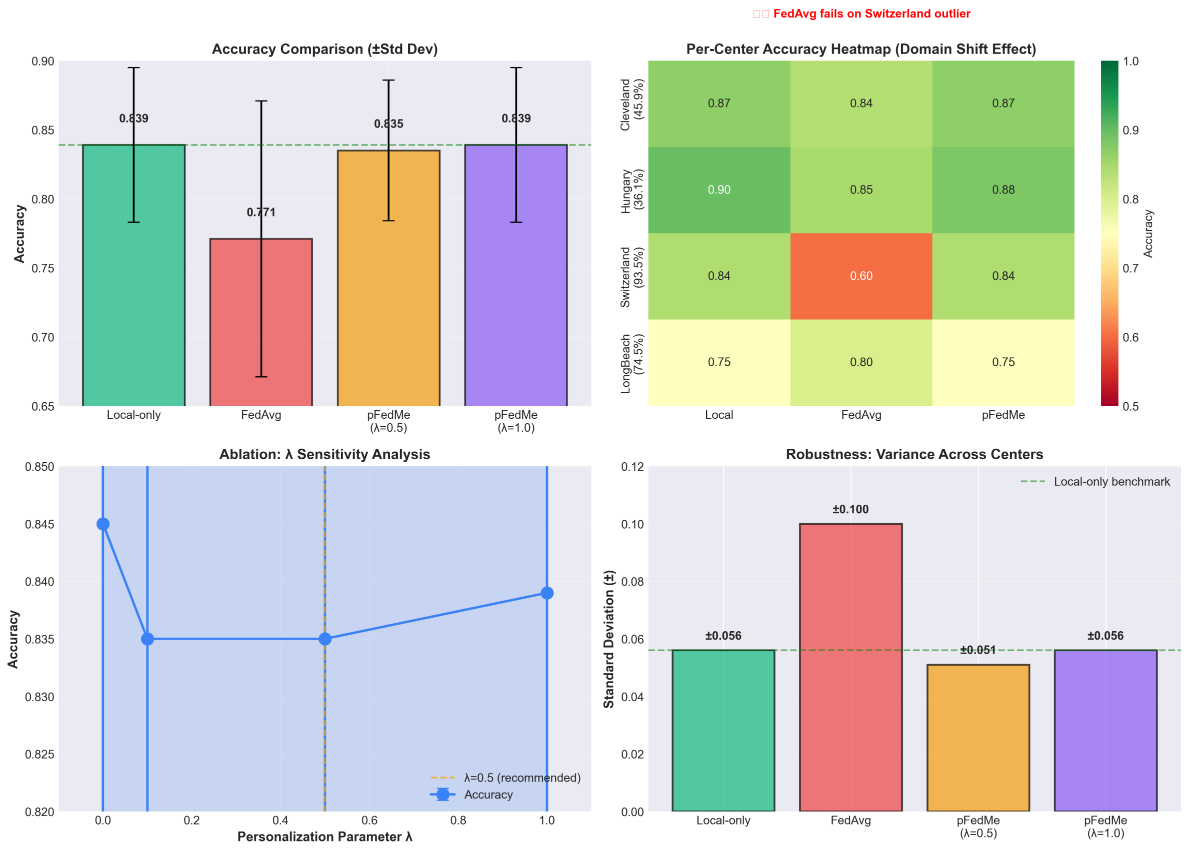

Figure 5: Critical analysis and deployment-oriented summary from W4.

5.1 Limitations and Future Work

Table 6: Limitations and future work.

Area	Current limitation	Future work	Priority
Statistical power	Only four centers are available.	Repeat the protocol on larger multi-hospital datasets and report confidence intervals.	High
Model capacity	Logistic regression may underfit nonlinear relationships.	Compare neural and tree-based models under the same FL protocol.	Medium
Communication realism	Bandwidth, latency, and drop-out are not measured.	Add communication-cost logging and partial participation.	High
Clinical validation	Evaluation is offline.	Design external validation and inspect calibration.	High
Privacy and governance	Secure aggregation and differential privacy are not implemented.	Prototype privacy-preserving aggregation and governance workflows.	Critical

6 Reproducibility

The project is organized around notebooks and reusable source modules. The main implementation files are `src/fedavg.py`, `src/pfedme.py`, `src/evaluation.py`, and `data/data_pipeline.py`.

The main notebooks are `W2_data_exploration.ipynb`, `W3_federated_learning.ipynb`, and `W4_critical_analysis.ipynb`.

7 Team Contributions

Table 7: Team contribution summary.

Member	Contribution area
HADJADJ Daoud	FedAvg/pFedMe implementation, experiment execution, evaluation.
KHODJA M'hand Fassih	Data preprocessing, visualization, critical analysis.
KOUTCHOUK Abderrahmane	Statistical significance analysis and per-center diagnostics.
OUADI Chaima	Report writing, limitations, and future-work refinement.
BOUBAHA Rosa	Results verification, table consistency, and reproducibility checks.
RAHAL Nadjiba	Literature alignment, deployment discussion, and final review.

8 Conclusion

This project demonstrates that personalization is a practical tool for federated learning under healthcare domain shift. FedAvg reaches 0.775 average accuracy, while pFedMe reaches 0.835 and avoids the major Switzerland-center failure observed with the global model. The results support personalized FL for non-IID medical data, while larger datasets, privacy-preserving aggregation, communication-aware evaluation, and clinical validation remain important next steps.

References

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